

14 Homeostasis

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Negative feedback detects deviation from a set point and produces a response opposing the change. Pancreatic β cells release insulin at high blood glucose, increasing uptake and glycogen synthesis; α cells release glucagon at low glucose, promoting glycogen breakdown and gluconeogenesis. Kidney nephrons filter blood and selectively reabsorb useful solutes; ADH increases water permeability of collecting ducts.

Worked example: follow the reasoning

After dehydration, plasma water potential falls, ADH rises and more water is reabsorbed; urine becomes more concentrated and lower in volume.

Quick check · multiple choice

A Which hormone usually increases after blood glucose rises?

[1]

- A Glucagon
- B Insulin
- C ADH always
- D Adrenaline only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain the response of insulin and glucagon after a carbohydrate-rich meal.

[3]

Answers and explanation

Quick check: B. Insulin

Question 1 [3 marks]

Blood glucose increases [1]; insulin increases and promotes cellular uptake/storage as glycogen [1]; glucagon secretion decreases [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.